AMUTHEEZAN SIVAGNANAM

Visa Status: Lawful Permanent Resident / Green Card (USA), Citizenship: SRI LANKA
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SUMMARY

PhD Candidate and **Machine Learning Researcher** specializing in **Deep Reinforcement Learning** with **six** years of academic research and **two** years of industrial software development experience. Expertise in **scalable deep learning and reinforcement learning systems**, ML **training, and deployment**, complemented by a **strong publication record (h-index 6)**. Proficient in **Python**, **TensorFlow**, **PyTorch**, AWS, Agile and CI/CD practices, with hands-on experience in **optimizing ML models and pipelines**.

EDUCATION

Pennsylvania State University
Ph.D., Computer Science — *Aug 2025*
Dissertation: *Application of Deep Reinforcement Learning to Solve Optimization Problems in Transportation Domains*

University of Houston
M.S., Computer Science — *Aug 2022*

University of Moratuwa
B.S., (Hons) Engineering (Computer Science and Engineering) — *Jan 2018*
Final Year Project: *Sentimental Analysis of Twitter using Semi-Supervised Approaches*

AREAS OF EXPERTISE

Reinforcement Learning, Optimization, Operational Research

SKILLS

Programming Languages: Python, C++, Java, C, JavaScript, PHP, MATLAB
Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Ray, RLLib, OpenAI Gym,

NumPy, pandas, matplotlib, CPLEX, Gurobi, Mosek, OR-Tools
Databases: SQL, MySQL, SQLite, MongoDB
Tools & Platforms: Amazon Web Services, Git, GitHub, Linux, Docker, Spark, CI/CD, Bash
Methodologies: Object-Oriented Development, Agile, Behaviour Driven Development

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RESEARCH EXPERIENCE

Pennsylvania State University — Graduate Research Assistant
Applied Artificial Intelligence Lab
Aug. 2022 - May 2025

- Spearheaded **research projects** funded by the **U.S. Department of Energy (DOE)** and the **National Science Foundation (NSF)**, leading to advancements in **cost- and energy-efficient technologies**
- Studied **real-world decision-making problems** and identified **gaps** in existing solution approaches, leading to the development of **innovative strategies** for improved **decision-making**
- Developed **mathematical models** and formulated **problems** to address these challenges, resulting in more **accurate predictions** and solutions for **complex issues**
- Applied **artificial intelligence–based solutions** to tackle **real-world problems** and successfully **deployed** them in relevant **industries**
- Published **research findings** in **AI conferences** (**ICML**) and assisted in preparing **slides** for presenting results at **DOE** and **NSF meetings**
- Developed a **deep reinforcement learning (DRL)** method (using **DDPG**) for **proactive responder repositioning** in **emergency management**, achieving significantly **faster response times** while reducing **operational delays**
- Applied **DRL** (using **DQN**) to solve **online vehicle routing with advance booking**, enabling **real-time confirmations** within seconds

The University of Houston — Graduate Research Assistant
Resilient Networks and Systems Lab
Sep. 2019 - Aug. 2022

- Spearheaded **research projects** funded by the **U.S. Department of Energy (DOE)** and the **National Science Foundation (NSF)**, leading to advancements in **energy-efficient technologies**
- Studied **real-world decision-making problems** and identified **gaps** in existing solution approaches, leading to the development of **novel methodologies** for enhanced **decision-making**
- Developed **mathematical models** and formulated **problem statements** to effectively address these challenges, resulting in more **accurate predictions** and solutions for **complex issues**

- Applied **AI-based solution approaches** to **real-world problems** and successfully **deployed** them in relevant **industries**
- Published **research findings** in **AI conferences** (**AAAI**, **IJCAI**) and assisted preparing **slides** for presenting results at **DOE** and **NSF meetings**
- Introduced **heuristics** that reduced **annual energy costs** by **$140K** for **public transit agencies** operating **mixed fleets of buses**
- Implemented a **deep reinforcement learning** (using **DQN**) approach that reduced **operational costs** by **20%** by enabling **online booking** for traditionally offline **Vehicle Routing Problems (VRPs)**
- Collected and analyzed **real-world data** using **Python** to identify **trends**, such as changes in **paratransit operations** before and after **COVID-19** and the impact of **vulnerability reward programs**

SOFTWARE ENGINEERING EXPERIENCE

LSEG Technology — Software Engineer

Post Trade Team

Jan 2018 – Jul 2019

- Developed application software using **Object-Oriented principles**, enhancing **code maintainability** and **scalability**
- Implemented **agile development practices**, such as **Scrum**, to improve **team collaboration** and **project delivery speed**
- Developed software solutions using **Java**, **Python**, and **C++**, improving **system performance** and **reliability**
- Introduced **Unit Testing** for **Libraries** in **Post Trade C++ Code**
- Modified **database structures** and tested using **BDD-based testing**, ensuring **data integrity** and **system functionality**
- Identified and replaced **duplicated error codes** with valid new error codes, improving **system reliability**
- Developed a **dynamic scripting object (DSO)** for **front-end end-to-end testing** using **Java**, enhancing **test coverage** and **efficiency**
- Analyzed and modified **back-end regression scripts** to work with both old and new **testing frameworks**, ensuring **compatibility** and reducing **testing time**
- Developed a new **report generation system** for the testing framework, **automating email dispatch** at the end of regression and improving **communication efficiency**
- Completed **code integration tasks** related to **back-end regression**, streamlining the **development process** and enhancing **system performance**
- Updated **automatic updates** to **auto-generated codes** based on **Database changes** using **Integration Plans**
- Contributed to **code integration** and **deployment plans**, ensuring **seamless software updates** and minimizing **downtime**
- Participated in **professional training programs** conducted by **Millennium IT Software** and **Post Trade Team**

- Worked on **Front-End Development** for both **Product** and **Solution** which consists of **Enhancement**, **Bug Fixing**, **Merging** and **Introducing new features**

WSO2 Lanka PVT Ltd — Software Engineering Intern
Data Analytics Team
Jul 2016 – Dec 2016

- Developed an HL7 Monitoring Solution by integrating **WSO2 ESB**, **DAS**, and **BAM**, enabling real-time and batch processing of healthcare data
- Developed an alert generation system analyzing descriptive **HL7/FHIR** data to monitor **disease outbreaks**, triggering timely email and SMS notifications
- Created **Spark scripts** for batch analytics and **Siddhi execution plans** for real-time alerts, including disease outbreak and patient wait-time notifications
- Engineered a mechanism to evaluate **hospital functionality** by assessing **admission and discharge** messages, including bed and oxygen cylinder availability
- Implemented interactive dashboards using **Jaggery**, **JavaScript**, **jQuery**, **Leaflet.js**, and **DataTables**, featuring gadgets like charts, maps, and tables
- Utilized **HAPI library and test panel** to simulate HL7 v2 messages and validate the monitoring pipeline end-to-end
- Packaged the entire monitoring solution as a **Carbon Application (CApp)**, bundling event streams, receivers, publishers, and visualization artifacts
- Attended workshops and gained hands-on experience with **Git**, **MSF4J (Microservices for Java)**, and **WSO2 product architecture** for enterprise middleware solutions

PUBLICATIONS

CONFERENCE PROCEEDINGS

[C7] **Sivagnanam, A.,** Pettet, A., Lee, H., Mukhopadhyay, A., Dubey, A., & Laszka, A. (2024). Multi-agent reinforcement learning with hierarchical coordination for emergency responder stationing. In *Proceedings of the International Conference on Machine Learning (ICML-24)*.

[C6] R. Sen, **A. Sivagnanam,** A. Laszka, A. Mukhopadhyay and A. Dubey, "Grid-Aware Charging and Operational Optimization for Mixed-Fleet Public Transit," *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*, Edmonton, AB, Canada, 2024, pp. 4172-4179.

[C5] Pavia, S., Rogers, D., **Sivagnanam, A.**, Wilbur, M., Edirimanna, D., Kim, Y., Mukhopadhyay, A., Pugliese, P., Samaranayake, S., Laszka, A., & Dubey, A. (2024). SmartTransit.AI: A dynamic paratransit and microtransit application. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24)* (pp. 8767–8770).

[C4] Atefi, S., **Sivagnanam, A.,** Ayman, A., Grossklags, J., & Laszka, A. (2023, May). The benefits of vulnerability discovery and bug bounty programs: Case studies of Chromium and Firefox. In *Proceedings of the ACM Web Conference* (pp. 2209–2219).

[C3] **Sivagnanam, A**., Kadir, S. U., Mukhopadhyay, A., Pugliese, P., Dubey, A., Samaranayake, S., & Laszka, A. (2022, July). Offline vehicle routing problem with online bookings: A novel problem formulation with applications to paratransit. In *Proceedings of the Thirty-First International Joint Conference on Artificial Intelligence (IJCAI-22)* (pp. 3933–3939).

[C2] **Sivagnanam, A**., Ayman, A., Wilbur, M., Pugliese, P., Dubey, A., & Laszka, A. (2021, May). Minimizing energy use of mixed-fleet public transit for fixed-route service. In *Proceedings of the AAAI Conference on Artificial Intelligence (Vol. 35, No. 17, pp. 14930–14938).*

[C1] Ayman, A., Wilbur, M., **Sivagnanam, A.,** Pugliese, P., Dubey, A., & Laszka, A. (2020, September). *Data-driven prediction of route-level energy use for mixed-vehicle transit fleets.* In Proceedings of the 6th IEEE International Conference on Smart Computing (SMARTCOMP) (pp. 1–8). IEEE.

JOURNALS

[J2] Wilbur, M., Ayman, A., **Sivagnanam, A.,** Ouyang, A., Poon, V., Kabir, R., Vadali, A., Pugliese, P., Freudberg, D., Laszka, A., & Dubey, A. (2023). Impact of COVID-19 on Public Transit Accessibility and Ridership. *Transportation Research Record*, *2677*(4), 531-546.

[J1] Ayman, A., **Sivagnanam, A**., Wilbur, M., Pugliese, P., Dubey, A., & Laszka, A. (2022). Data-driven prediction and optimization of energy use for transit fleets of electric and ICE vehicles. *ACM Transactions on Internet Technology, 22*(1), Article 7, 1–29.

WORKSHOPS

[W1] **Sivagnanam, A.,** Atefi, S., Ayman, A., Grossklags, J., & Laszka, A. (2021). On the benefits of bug bounty programs: A study of Chromium vulnerabilities. In *Workshop on the Economics of Information Security (WEIS)* (Vol. 10).

BOOK CHAPTERS

[BC1] Wilbur, M., **Sivagnanam, A.,** Ayman, A., Samaranayake, S., Dubey, A., & Laszka, A. (2023, August). Artificial intelligence for smart transportation. In **Y. Vorobeychik & A. Mukhopadhyay (Eds.),** *Artificial Intelligence and Society* (book chapter). ACM Press.

TALKS

IJCAI (Vienna, Austria) 2022
For the paper: Offline Vehicle Routing Problem with Online Bookings:
A Novel Problem Formulation with Applications to Paratransit

WEIS (Virtual Workshop) 2021
For the workshop paper: On the Benefits of Bug Bounty Programs:
A Study of Chromium Vulnerabilities

AAAI (Virtual Conference) 2021
For the paper: Minimizing Energy Use of Mixed-Fleet Public Transit
for Fixed-Route Service

SERVICES

Program Committee Member
International Joint Conference on Artificial Intelligence (AI For Social Good) 2025

Reviewer
International Joint Conference on Artificial Intelligence (AI For Social Good) 2024

Reviewer
AI4Research Workshop @IJCAI2024 2024

Auxiliary Reviewer
22nd International Conference on Autonomous Agents and Multiagent Systems 2023

AWARDS AND HONORS

Information Security Quiz 2015 - Runners Up 2015
Issued by Sri Lanka Computer Emergency Readiness Team(SLCERT)
and Information and Communication Technology Agency of Sri Lanka(ICTA)

Mahapola Scholarship 2014 - 2017

REFERENCES

Dr. Aron Laszka (PhD Supervisor)
Position: **Assistant Professor**
Affiliation: **Pennsylvania State University**
Email: aql5923@psu.edu
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Website: https://aronlaszka.com

Dr. Abhishek Dubey
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Affiliation: **Vanderbilt University**
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Phone: **615-322-8775**
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Dr. Weidong Shi
Position: **Associate Professor**
Affiliation: **University of Houston**
Phone: **713-743-3045**
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